DAY 3

Shotgun Metagenomics & Assembly

QC & host removal • Kraken2 • MetaPhlAn4 • MEGAHIT/SPAdes • QUAST • SEQKIT

Amplicon vs Shotgun Metagenomics

Feature	Amplicon (16S)	Shotgun WGS
What is sequenced	One marker gene (16S rRNA)	All DNA in sample
PCR step	Yes — amplifies target gene	No — random fragmentation
Taxonomic resolution	Genus level (typically)	Species / strain level
Functional data*	None	Full gene complement
Genome reconstruction*	Not possible	Yes — MAGs
Novel organism detection	Limited by primers	Yes — unbiased
Cost (per sample)	~$30–80	~$100–300
Data size (per sample)	~10 MB	~1–10 GB
Bioinformatics complexity	Moderate	High
Best for*	Community profiling, large cohorts	Functional potential, MAGs, discovery

Today we focus on shotgun — the foundation for Day 4 (MAGs) and Day 5 (Functional Analysis).

Shotgun Metagenomics Workflow

01 Raw Reads QC

FASTQC + MultiQC

Check read quality, adapter contamination, base composition, and GC content.

02 Trimming

fastp / Trimmomatic

Remove adapters, low-quality bases; filter short reads.

03 Host Removal

Bowtie2 / BWA

Map reads to host genome (human/plant), keep unmapped reads.

04 Read-based Taxonomy

Kraken2 + Bracken

Fast k-mer classification. Bracken estimates species-level abundances.

05 De Novo Assembly

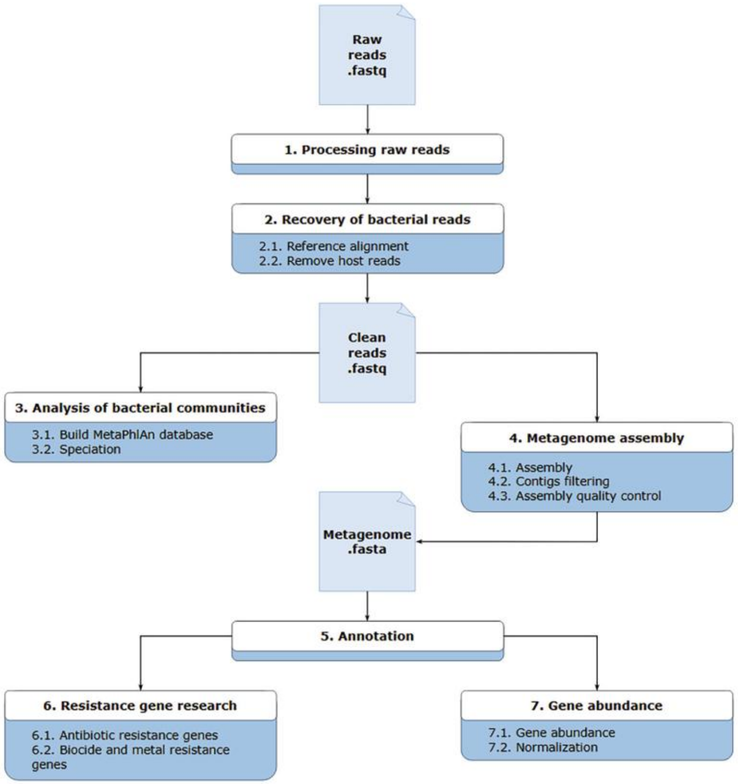
MEGAHIT / SPAdes

Assemble reads into contigs using de Bruijn graphs.

06 Assembly QC

QUAST + SEQKIT

Evaluate contig statistics: N50, length distribution, genome completeness.



Raw reads .fastq
1. Processing raw reads
2. Recovery of bacterial reads
2.1. Reference alignment
2.2. Remove host reads
Clean reads .fastq
3. Analysis of bacterial communities
3.1. Build MetaPhlAn database
3.2. Speciation
4. Metagenome assembly
4.1. Assembly
4.2. Contigs filtering
4.3. Assembly quality control
Metagenome .fasta
5. Annotation
6. Resistance gene research
6.1. Antibiotic resistance genes
6.2. Biocide and metal resistance genes
7. Gene abundance
7.1. Gene abundance
7.2. Normalization

Today's Dataset

Category	This course's data	Note
Study	TransplantLines gut metagenome (PRJNA1047900)	Zhang et al. 2024, mSystems
Samples analysed today	6 total: 3 Lung Transplant Recipient (LTR) + 3 HC	Same 6 samples, both Kraken2/Bracken and MetaPhlAn4
Subsampling depth	1,000,000 read pairs per sample	Uniform across all 6 — fair group comparison
Full-depth copies (not used today)	SRR27027652 (LTR), SRR27027756 (HC)	Set aside for homework / Day 5 mini-project
Soil sample	None in Day 3	Reserved dataset (PRJDB18822) for the mini-project — don't peek!

All 6 samples are gut — see data-README.md for full provenance and the subsampling rationale.

SECTION 1

QC & Preprocessing

FASTQC • MultiQC • fastp • Host Removal

Quality Control: FASTQC & MultiQC

FASTQC

Per-sample QC

Per-base quality

Phred scores along read length — look for quality drop at 3' end

Per-sequence quality

Distribution of mean quality per read

GC content

Should be approximately normal; spikes indicate contamination

Sequence duplication

High duplication → overamplification or low library complexity

Adapter content

Detect Illumina adapters that must be trimmed before analysis

Overrepresented sequences

Adapter dimers, ribosomal RNA, or contaminants

```
fastqc sample_R1.fastq.gz sample_R2.fastq.gz -o qc_output/
```

MultiQC

Aggregate across samples

- Aggregates FASTQC reports from all samples into one interactive HTML report
- Works with 100+ bioinformatics tools: FASTQC, Trimmomatic, Bowtie2, STAR, QUAST, Kraken2...
- Side-by-side comparison of quality metrics across many samples
- Instantly identify outlier samples (low quality, wrong GC content, etc.)
- Generates publication-quality figures (barplots, heatmaps, line plots)
- Supports custom modules via plugin system

```
multiqc qc_output/ -o multiqc_report/
```

Trimming & Host DNA Removal

fastp

Fast & feature-rich trimmer

- Adapter auto-detection (no need to specify adapter sequences)
- Quality trimming: --qualified_quality_phred 20
- Minimum read length: --length_required 50
- Deduplication: --dedup
- JSON + HTML report generated automatically
- Much faster than Trimmomatic (multi-threaded)

```
fastp -i R1.fastq.gz -I R2.fastq.gz \
  -o R1_trimmed.fastq.gz -O R2_trimmed.fastq.gz \
  -q 20 -l 50 --detect_adapter_for_pe -j fastp.json
```

Alternative trimmer: Trimmomatic (ILLUMINACLIP:adapters.fa:2:30:10 LEADING:3 TRAILING:3 SLIDINGWINDOW:4:15 MINLEN:36)
Beyond the host: reagent and lab-derived contamination (the "kitome") can dominate low-biomass samples — always include extraction/negative controls (Ch.2).

Host DNA Removal

Why? Human samples may contain 50–90% host DNA.
Without removal, host reads dominate assembly and classification, wasting compute and distorting results.

Strategy: Bowtie2 / BWA

1. Build index of host genome (GRCh38 for human)
2. Map reads to host: Bowtie2 --very-sensitive
3. Extract UNMAPPED reads → microbial reads
4. Convert SAM → BAM → filter unmapped

```
bowtie2 -x GRCh38 -1 R1.fastq.gz -2 R2.fastq.gz \
  --un-conc-gz clean_%.fastq.gz \
  -p 8 --very-sensitive | samtools view -bS > host.bam

# clean_1.fastq.gz and clean_2.fastq.gz = microbial reads
```

SECTION 2

Read-Based Taxonomy

Kraken2 • Bracken • MetaPhlAn4

Kraken2 & Bracken: k-mer Taxonomic Classification

How Kraken2 Works

1. Database build

k-mer extracted from all reference genomes, stored in hash table with taxon IDs

2. Read classification

Each read split into k-mers (default k=35). Each k-mer queried against database.

3. LCA assignment

Lowest Common Ancestor of all matching k-mer taxa assigned to read. Handles multi-mapping.

4. Output

Per-read classification + report with counts at each taxonomic level.

```
kraken2 --db /path/to/kraken2_db --paired R1.fastq.gz R2.fastq.gz --report kraken2.report --output kraken2.out
bracken -d kraken2_db -i kraken2.report -o bracken.txt -r 150 -l S
```

Bracken

Bayesian Re-estimation of Abundance with KrakEN

Kraken2 classifies reads but gives counts, not true abundances. Bracken uses Bayesian re-estimation to provide accurate species-level abundance estimates from genus/species-level counts.

MetaPhlAn4

Marker gene based profiling. Uses ~5.1M clade-specific marker genes (not k-mers). More sensitive for novel species. Requires smaller database than Kraken2. Integrates with HUMAnN3.

SECTION 3

De Novo Assembly

MEGAHIT • SPAdes • Assembly QC

De Novo Assembly: Concepts

Goal: reconstruct longer genomic sequences (contigs) from short, overlapping reads — without a reference genome.

de Bruijn Graph (DBG)

Reads split into k-mers (overlapping subsequences of length k). Nodes = unique k-mers. Edges = overlaps. Contigs = paths through graph. Dominant approach for NGS.

k-mer Size

Smaller k: more connections, handles errors better, but more ambiguity.
Larger k: more specific, less ambiguity, but sensitive to errors.
MEGAHIT uses multiple k-mers iteratively.

Contigs vs Scaffolds

Contigs: continuous assembled sequences (no gaps).
Scaffolds: contigs joined using paired-end information with N-filled gaps.
For metagenomics, contigs ≥ 1000 bp are typically used downstream.

Coverage Depth

More reads → higher coverage → better assembly.
Recommended: ≥10× coverage per genome.
Abundant organisms assembled better than rare ones.
Very rare organisms may not assemble at all.

MEGAHIT vs metaSPAdes

Feature	MEGAHIT	metaSPAdes
Speed	★★★★★ Very fast (minutes–hours)	★★★☆☆ Slow (hours–days)
Memory	★★★★★ Low (8–32 GB typical)	★★☆☆☆ High (100+ GB for complex)
Assembly quality	★★★☆☆ Good	★★★★★ Better for complex/low cov.
k-mer strategy	Multiple k-mers iteratively	Multiple k-mers simultaneously
Long reads support	No	Yes (hybrid assembly)
Best for	Large datasets, resource-limited, standard QC	Complex communities, higher quality assemblies
Command	`megahit -1 R1.fq.gz -2 R2.fq.gz -o megahit_out`	`spades.py --meta -1 R1.fq.gz -2 R2.fq.gz -o spades_out`

For this course we use MEGAHIT (speed + memory). Use metaSPAdes for publication-quality assemblies.

Assembly QC: QUAST & SEQKIT

QUAST / MetaQUAST

Metric	Direction	Description
Total length		Sum of all contig lengths — estimate of total metagenome size sequenced
# Contigs	lower	Total assembled sequences. Fewer contigs = more complete assembly
N50	higher	Contig length at which 50% of the assembly is in contigs of this length or longer
L50	lower	Fewest contigs whose combined length covers 50% of assembly. Fewer = better
Largest contig	higher	Longest single sequence — indicator of assembly continuity
GC content		Should match expected GC% for the community; flags contamination

```
quast.py assembly.fasta -o quast_output --threads 8
```

SEQKIT

Swiss-army knife for sequence files (FASTA/FASTQ).

`seqkit stats`
Summary: length, N50, count

`seqkit grep`
Extract sequences by name/pattern

`seqkit sort`
Sort by length (longest first)

`seqkit seq`
Filter contigs ≥ 1000 bp

`seqkit sample`
Subsample 10% of reads

`seqkit fx2tab`
Convert to tabular (name, len, GC)

AFTERNOON PRACTICAL

Today's Hands-On Session

1 QC & Trimming

Run FASTQC on shotgun reads. Check quality with MultiQC. Trim adapters with fastp. Compare before/after QC stats with SEQKIT.

2 Host Read Removal

Remove human reads using Bowtie2 against GRCh38. Count reads before/after. Inspect % host contamination across soil vs gut samples.

3 Taxonomic Classification

Run Kraken2 on cleaned reads. Apply Bracken for species-level abundance. Compare soil vs gut microbiome profiles.

4 De Novo Assembly

Assemble cleaned reads with MEGAHIT. Run QUAST for assembly statistics. Filter contigs ≥ 1000 bp with SEQKIT. Compare assemblies.

Day 3 Summary

Amplicon vs Shotgun	Shotgun sequences ALL DNA — gives taxonomy AND functional potential + enables MAG reconstruction.
QC	FASTQC identifies read quality issues; fastp trims adapters and low-quality bases efficiently.
Host removal	Map reads to host genome with Bowtie2; retain only unmapped (microbial) reads for downstream analysis.
Kraken2 + Bracken	k-mer classification provides fast taxonomy; Bracken re-estimates species-level abundance.
MEGAHIT	Fast, memory-efficient assembler using iterative de Bruijn graphs. Standard for metagenomics.
QUAST + SEQKIT	Evaluate assemblies (N50, contig count); manipulate FASTA/FASTQ with flexible SEQKIT toolkit.

Day 4 tomorrow → MAGs: Binning (MetaBAT2) → Quality (CheckM2) → Taxonomy (GTDB-Tk)

KEY REFERENCES

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